

Simulation-informed auxiliary learning improves low-data nanobody thermal stability prediction

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Molecular simulation can generate protein-property labels at a scale that is difficult to achieve experimentally, but those labels often describe physical quantities that are related to, rather than identical with, the experimental target. We study this problem in low-data nanobody thermal-stability prediction: can simulation-derived labels improve melting-temperature (T_m) prediction when the most informative source labels are mutation free energies rather than T_m measurements? We developed a target-centered auxiliary-learning framework in which experimental T_m prediction is the target task and computational labels are source tasks sharing the same ESM2 sequence encoder. All model selection was performed using an experimental development set, and all claims were evaluated on the same held-out experimental T_m test set. Free-energy perturbation (FEP) mutation-effect labels provided the clearest improvement, reducing held-out T_m mean absolute error from 6.61 to 6.26°C relative to T_m -only training (paired improvement, 0.35°C; 90% confidence interval, 0.24–0.46°C). The effect was source-dependent: Rosetta-scored generated variants produced a weaker but positive signal, whereas an MD-derived contact-persistence label did not improve test performance. Encoder-size controls showed that increasing ESM2 from 8M to 35M or 650M parameters did not explain the gain. These results support a selective view of simulation-to-experiment transfer: useful auxiliary labels are not merely additional data, but encode physical information that transfers to the experimental phenotype. In nanobody T_m prediction, mutation free-energy labels appear to complement sparse absolute T_m anchors by providing local stability-change information over sequence perturbations.

nanobody thermostability | melting temperature | simulation-to-real transfer learning | protein language model | free energy perturbation
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Introduction

Machine learning has made it possible to screen large molecular and materials spaces before costly experimental testing, but its usefulness still depends on the availability of reliable property labels. In materials informatics, this data bottleneck has motivated a simulation-to-real (Sim2Real) strategy: large computational databases generated by first-principles calculations, quantum chemistry, or molecular dynamics simulations are used as source data, and the resulting models are transferred to smaller experimental target datasets (1–3). A recent study in polymers and inorganic materials showed that the downstream prediction error of

transferred models can scale with the amount of computational data, providing a quantitative way to judge whether a growing simulation database has real experimental value (3). This framing is important because simulation data are not useful by default: simulations and experiments are separated by model assumptions, sampling limits, and systematic domain gaps.

Protein engineering faces an analogous problem, but with an additional layer of physical mismatch. Experimental measurements of protein properties are expensive and sparse, whereas biomolecular simulations and physics-based calculations can generate labels for many designed or mutated sequences. The challenge is sharper than in many property-matched transfer settings, because the simulated quantity may not be the same as the experimental target. For example, a target phenotype may be the melting temperature (T_m) of a protein, while a tractable simulation may report a mutation free-energy change, a structure-derived score, or a trajectory summary. The central question is therefore not simply whether simulation data can be added to a training set, but which simulated physical quantities can transfer to the experimental phenotype.

We examine this question using nanobody thermostability as a concrete case. Nanobodies, or single-domain VHH antibodies, are attractive engineering scaffolds because of their small size, modularity, and binding properties (4, 5). Their practical use, however, depends on thermal stability during expression, storage, formulation, and downstream screening. Experimental T_m measurements remain limited, and direct molecular simulation of thermal stability is computationally demanding (6). Sequence-based predictors and protein language models (PLMs) offer an attractive alternative: PLMs such as ESM2 learn broadly useful protein representations from large sequence corpora (7, 8), and recent methods have used deep sequence representations to predict protein or nanobody melting temperatures (9–11). Nevertheless, published benchmarks and related nanobody T_m studies indicate that target-specific supervision and careful validation remain essential (12–14).

Here we ask whether simulation-derived labels can provide that additional supervision when experimental T_m labels are scarce. We formulate T_m prediction as the target task and train auxiliary source tasks from computational labels using a shared ESM2 sequence encoder.

The auxiliary labels include FEP-derived mutation free energies, Rosetta and ThermoMPNN mutation-effect labels, Rosetta scores on generated and random variants, and an MD-derived contact-persistence label. To keep the comparison target-centered, all model settings are selected using an experimental development set, and all final claims are evaluated on the same held-out experimental Tm test set with paired error comparisons. This design prevents high performance on an auxiliary task from being mistaken for improved Tm prediction.

The main result is that the benefit of simulation-derived supervision is highly source-dependent. FEP mutation free-energy labels provide the strongest improvement over Tm-only training, whereas the MD-derived contact-persistence label does not improve held-out Tm prediction under the same evaluation design. Rosetta-scored generated variants provide a weaker but positive signal, suggesting a connection to future generator-physics-predictor design loops. Taken together, these results extend the Sim2Real argument from materials informatics to a biomolecular setting in which the source and target labels differ in physical meaning. Useful simulation data are not merely abundant; they encode a physical quantity that transfers to the experimental property of interest.

Methods

Experimental Tm target data. Nanobody melting-temperature (Tm) prediction was formulated as a supervised sequence-regression task. Experimental Tm measurements defined the target endpoint for model selection and final evaluation. The data were split into training, development, and held-out test sets before model comparison. The development set was used for selecting model settings, and the held-out test set was used only for the final paired comparisons reported in the Results.

Computational source labels. Computational labels were treated as auxiliary source tasks rather than as additional Tm measurements. The source-label collection included mutation free-energy labels from free-energy perturbation calculations, structure-based mutation-effect labels, scores assigned to generated and random variants, and an MD-derived contact-persistence label. Label-count experiments were performed by training models with increasing numbers of source labels while preserving the same target-task evaluation protocol.

Auxiliary-learning model. All models used an ESM2 sequence encoder followed by task-specific prediction heads. In the Tm-only baseline, the encoder and Tm head were trained using only experimental Tm labels. In auxiliary-learning models, the Tm head and a source-label head shared the sequence encoder, allowing the computational source task to influence the representation. Encoder-size controls compared the same protocol across ESM2 model sizes. Frozen-encoder controls kept

the pretrained sequence representation fixed and trained only the task heads.

Model selection and evaluation. To avoid selecting models by auxiliary-task performance, all candidate settings were chosen using the experimental Tm development set. Final performance was reported as mean absolute error on the held-out experimental Tm test set. Auxiliary sources were compared against the Tm-only baseline by paired differences in absolute error on the same test examples. Confidence intervals were estimated by bootstrap resampling of held-out test examples.

Results

A target-centered framework for using simulation labels in Tm prediction. We first established an evaluation framework for asking whether simulation-derived labels improve experimental Tm prediction. The target task was nanobody Tm regression from sequence. Computational labels were treated as auxiliary source tasks rather than as substitutes for experimental Tm. Each model shared an ESM2 sequence encoder between a Tm prediction head and an auxiliary source-label head, allowing the source task to shape the representation while keeping the final objective anchored to experimental Tm prediction (Fig. 1).

The comparison was deliberately target-centered. All candidate settings were selected using an experimental Tm development set, and the final numbers were computed only on a held-out experimental Tm test set. Improvements were assessed by paired differences in absolute error on the same test examples. This design addresses a common ambiguity in simulation-to-experiment learning: a source task can be learnable and physically meaningful without improving the target experimental phenotype.

Auxiliary-label scaling depends on the physical source label. We next varied the number of experimental and computational labels to determine whether more source data translated into better Tm prediction. As expected, increasing the number of experimental Tm labels improved Tm-only performance (Fig. 2). FEP mutation free-energy labels also showed a beneficial trend: the full FEP label set gave the best held-out Tm error among the FEP label-count settings. The curve was not strictly monotonic, so we do not interpret it as a clean power law. Instead, we interpret it as evidence that additional FEP labels can improve the target task when they are selected and evaluated through the experimental Tm endpoint.

The MD-derived contact-persistence label behaved differently. Increasing the number of these labels did not yield a robust improvement over Tm-only learning, even though the source labels were also generated from simulations. Thus, the label-count experiment already suggested the central result of the paper: auxiliary data

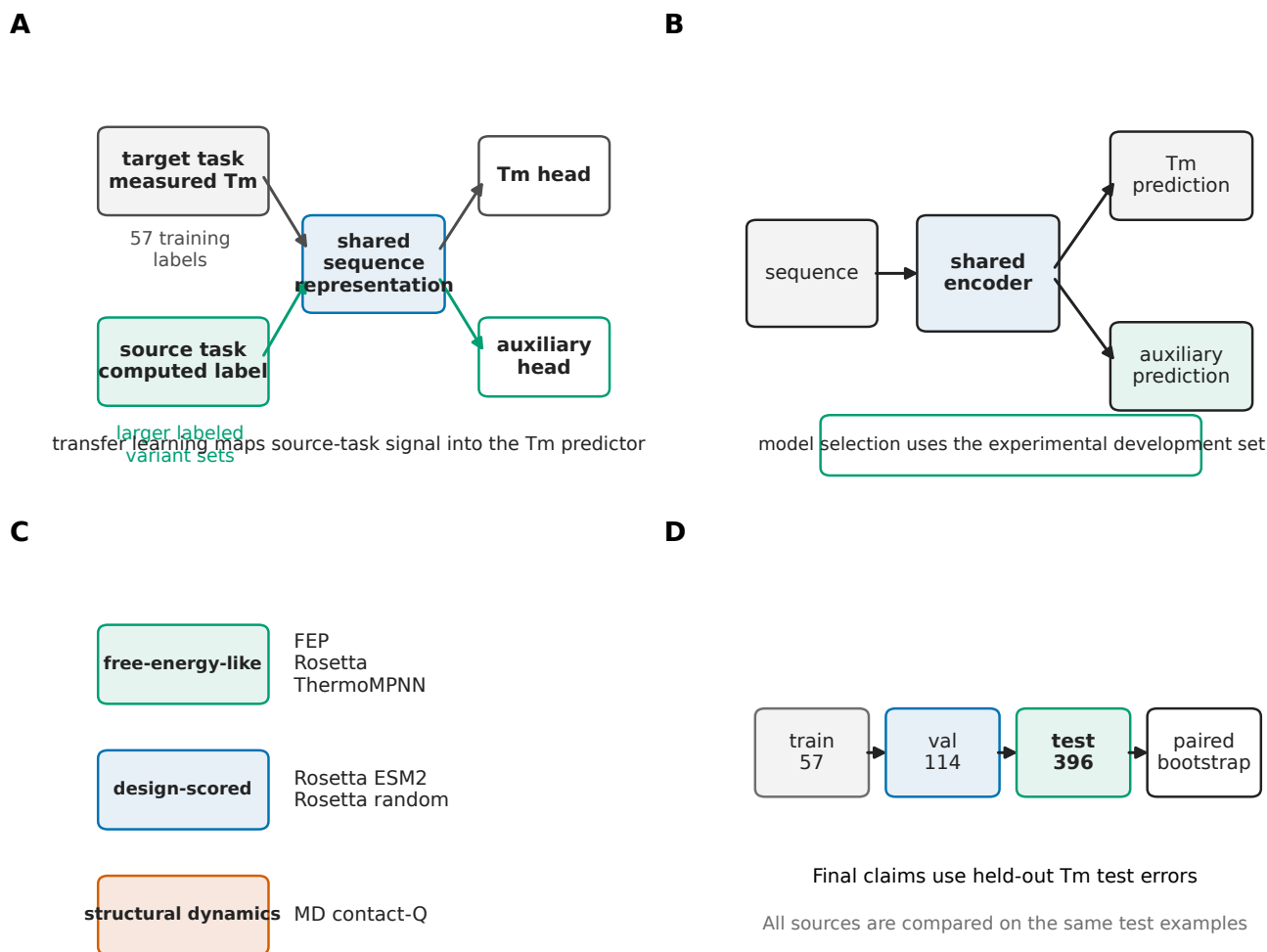


Fig. 1. Auxiliary-learning framework for using simulation data in experimental Tm prediction. (a) Experimental Tm labels are scarce, whereas simulation and physics-based calculations can provide larger source-label pools. (b) Nanobody Tm prediction is used as a concrete case in which the source labels are related to, but not identical with, the experimental target. (c) The model shares an ESM2 sequence encoder between the Tm target head and an auxiliary source-label head. (d) Model settings are selected on an experimental development set, and final claims are evaluated on held-out experimental Tm test examples using paired error statistics.

quantity is not sufficient. The transferable physical content of the source label matters.

FEP mutation free-energy labels provide the strongest final improvement. We then compared the selected settings for each auxiliary source. Tm-only training achieved a held-out test MAE of 6.61°C. Adding FEP mutation free-energy labels reduced the MAE to 6.26°C, corresponding to a paired improvement of 0.35°C (90% confidence interval, 0.24–0.46°C; Fig. 3). This was the clearest improvement observed across all source labels. The other sources showed weaker or negative effects. Rosetta-scored variants generated by ESM2 improved over Tm-only learning (paired improvement, 0.17°C; 90% confidence interval, 0.03–0.31°C), and ThermoMPNN mutation-effect labels showed a similar but smaller gain. Rosetta mutation scores and Rosetta-scored random variants were weaker and had confidence intervals that crossed zero. The MD-derived contact-

persistence label worsened the final test error relative to Tm-only learning. These results support a selective, source-dependent interpretation: mutation free-energy-like labels were useful, but a structural trajectory summary was not sufficient for this target task.

The FEP gain is not explained by encoder size alone. Because PLM scale can affect downstream performance, we tested whether the benefit of FEP labels could be reproduced simply by increasing the ESM2 encoder size. It could not. In the Tm-only setting, larger 35M and 650M encoders did not improve held-out performance relative to the 8M encoder. In the FEP setting, the 35M and 650M encoders reached 6.50 and 6.65°C MAE, respectively, both worse than the 8M FEP model. FEP labels remained beneficial within each size comparison, but the best absolute performance was obtained with the smallest encoder and FEP auxiliary labels. Thus, in this low-Tm-data regime, the dominant gain came from

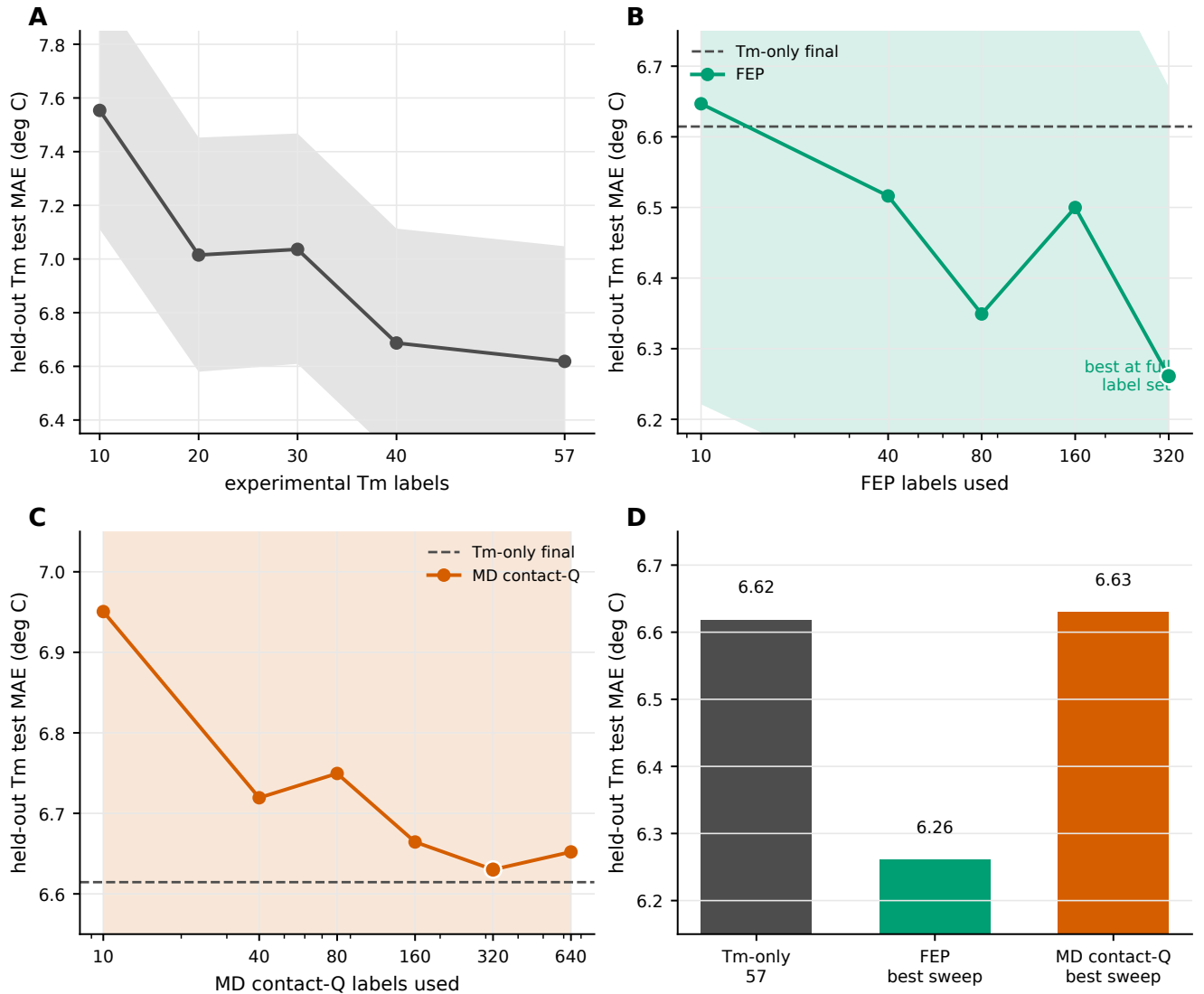


Fig. 2. Label-count scaling of experimental and computational supervision. (a) Tm-only learning improves as more experimental Tm labels are used. (b) FEP mutation free-energy labels reach their best held-out Tm error at the full source-label set, although the curve is not strictly monotonic. (c) The MD-derived contact-persistence label does not show a robust improvement over Tm-only learning. (d) Best points from the label-count sweeps separate FEP from the Tm-only and MD-derived contact-persistence references.

the physical source label rather than from scaling the language model.

Encoder adaptation helps exploit the auxiliary signal. We also asked whether the FEP gain required updating the sequence encoder. In a frozen-encoder control, overall errors increased, but FEP still improved over the frozen Tm-only baseline: the frozen Tm-only model achieved 7.38°C MAE, whereas frozen FEP achieved 7.20°C MAE. The improvement was smaller than in the updated-encoder setting, and the frozen FEP model remained worse than the updated-encoder FEP model. These results indicate that FEP labels carry useful information even when the representation is fixed, but limited encoder adaptation is important for extracting the strongest target-task benefit.

Physics-labeled generated variants connect auxiliary learning to design loops. Rosetta labels on ESM2-proposed variants improved Tm prediction over the Tm-only baseline, whereas the comparison between ESM2-proposed and random Rosetta-scored variants was suggestive but not conclusive. The paired improvement of ESM2-proposed variants over Tm-only learning was 0.17°C (90% confidence interval, 0.03–0.31°C). By contrast, the paired difference between ESM2-proposed and random variants was 0.07°C in favor of ESM2-proposed variants, with a confidence interval crossing zero. We therefore interpret this result conservatively. Physics-labeled generated variants retain useful supervised signal for Tm prediction, but the present data do not prove that the current generator is superior to random variant sampling.

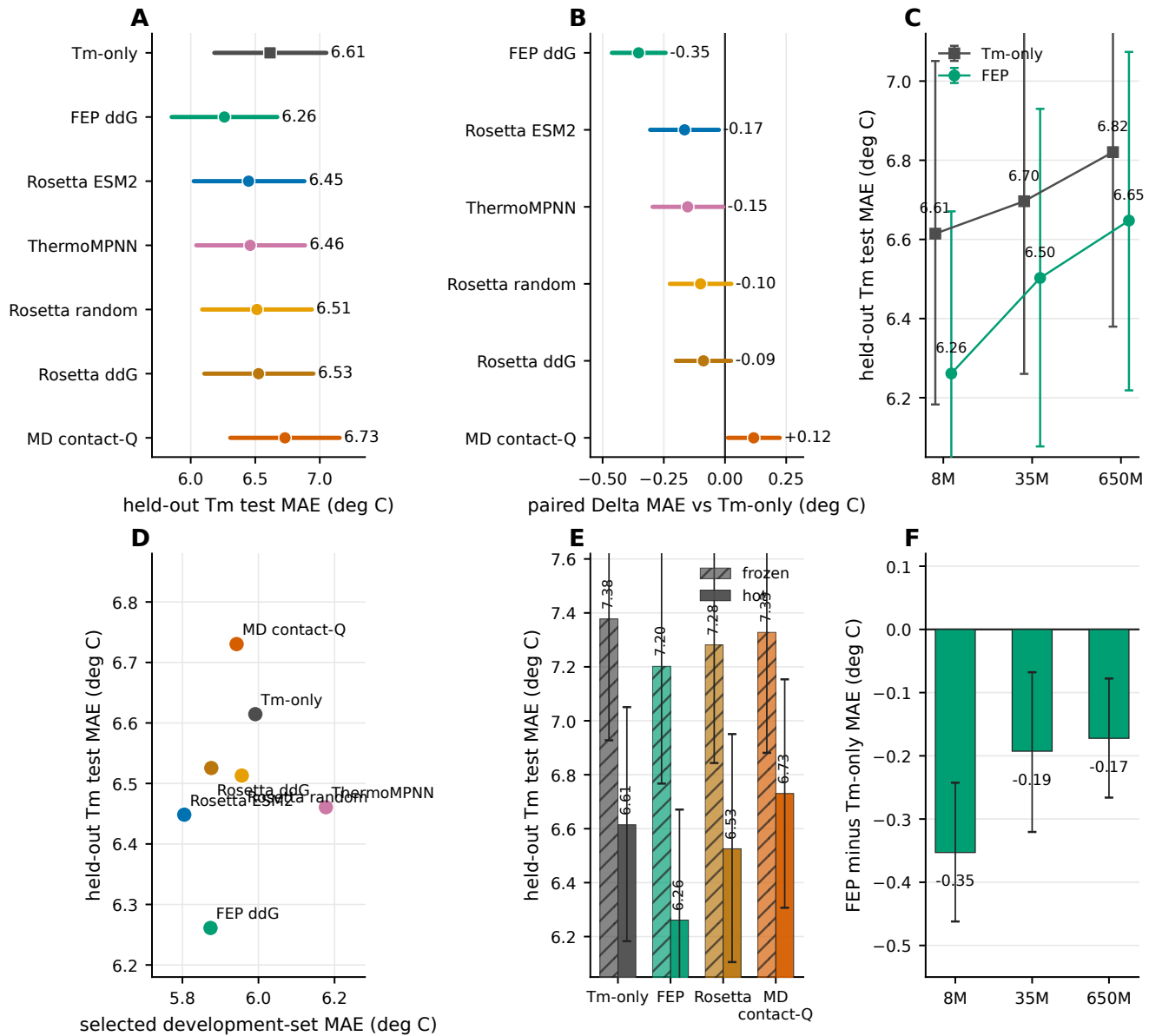


Fig. 3. Final held-out Tm performance across auxiliary sources and controls. (a) Final test MAE for Tm-only training and each auxiliary source. Intervals show bootstrap confidence intervals over test-set absolute errors. (b) Paired Δ MAE relative to Tm-only training. Negative values indicate improvement. (c) ESM2 size controls show that larger encoders do not improve the selected Tm-only or FEP settings. (d) Development-set performance and held-out test performance are related but not identical, emphasizing the need for final target-task testing. (e) Frozen- and updated-encoder comparisons show that FEP helps in both regimes, with the strongest absolute performance obtained when the sequence representation is updated. (f) Paired FEP gains remain negative across ESM2 sizes, but the best absolute error is obtained with the 8M encoder.

Mutation free-energy labels provide a physical interpretation. The final pattern suggests why FEP labels transfer to Tm prediction even though they are not Tm measurements. Experimental Tm labels provide sparse anchors for the absolute thermal-stability phenotype. Mutation free-energy labels provide local information about how stability changes under sequence perturbations. When learned jointly, these two views can shape a representation that is more useful for held-out Tm prediction than either sparse Tm labels alone or a generic simulation-derived summary. The negative result for the MD-derived contact-persistence label defines the bound-

ary of this interpretation: a simulation label must encode the right physical relation to the target phenotype.

Discussion

This study asked how simulation-derived labels should be used when the experimental target is scarce and the simulated quantity is related to, but not identical with, that target. Using nanobody Tm prediction as a case study, we found that FEP mutation free-energy labels improved held-out experimental Tm prediction more clearly than any other auxiliary source tested. The result was not reproduced by a simple MD-derived

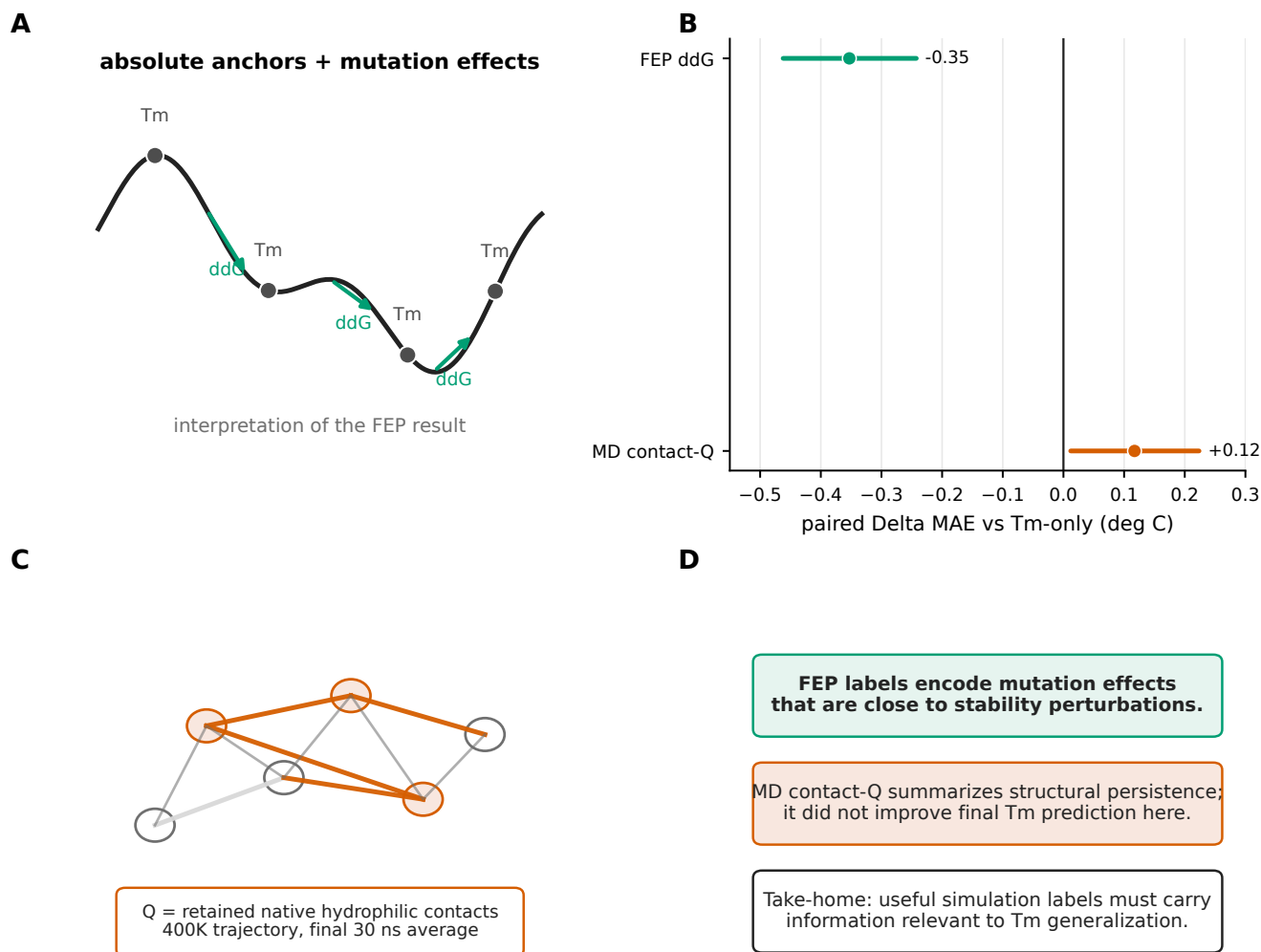


Fig. 4. Interpretation and boundary condition. (a) Experimental T_m labels provide sparse absolute stability anchors, whereas mutation free-energy labels provide local stability-change directions. (b) FEP improves T_m prediction, whereas the MD-derived contact-persistence label does not. (c) Contact persistence summarizes retained native contacts in high-temperature trajectories, but this structural-dynamics signal does not necessarily transfer to held-out T_m prediction. (d) The useful signal is not generic simulation augmentation, but auxiliary information with transferable physical content.

contact-persistence label, nor was it explained by using a larger ESM2 encoder. These findings support a selective view of Sim2Real learning for biomolecules: simulation data are useful when the source label encodes physical information that transfers to the experimental phenotype.

The main conceptual point is that the helpful source label did not measure T_m itself. FEP estimates mutation free-energy changes, whereas T_m is an experimental phenotype reflecting the temperature at which folded and unfolded states are balanced under a particular assay condition. These are not interchangeable quantities. Nevertheless, both describe the same underlying stability landscape from different perspectives. Sparse T_m measurements anchor absolute thermal-stability values for observed sequences, while mutation free-energy labels provide local directions in sequence space. The improvement from FEP therefore suggests that auxiliary

labels can help even when they are not direct surrogates for the target, provided that they describe a physically relevant perturbation of the target property.

This result extends the materials-informatics Sim2Real argument to a more heterogeneous biomolecular setting. In materials applications, transfer learning often links computational and experimental estimates of related material properties, and scaling behavior can quantify the downstream value of a growing computational database (3). In the present protein case, the most useful computational labels were not additional T_m measurements but mutation free energies. The label-count sweep for FEP was consistent with a beneficial scaling trend but was not smooth enough to claim a universal biomolecular scaling law. The stronger conclusion is source dependence: simulation-derived data should be evaluated by target-task performance, and source labels must be chosen for transferable physical content rather

than availability alone.

The encoder controls sharpen this conclusion. Prior work on PLMs has shown that fine-tuning can be critical for downstream protein-property prediction (14), and related nanobody T_m analyses suggest that larger pretrained ESM2 models do not automatically produce better T_m predictors. Our own size controls reached the same practical conclusion: 35M and 650M ESM2 encoders did not outperform the 8M encoder in either T_m-only or FEP-assisted settings. The best model used the 8M encoder with FEP auxiliary labels. Thus, for this dataset, the bottleneck was not simply language-model capacity. The more important ingredient was additional supervision carrying information about sequence-dependent stability changes.

The frozen-encoder experiment further indicates how the auxiliary signal is used. FEP labels improved performance even when the encoder was fixed, showing that part of the signal can be captured by the existing sequence representation. However, the best FEP model required updating the encoder. This suggests that FEP labels help most when they are allowed to reshape the representation toward stability-relevant sequence directions while still being selected by experimental T_m performance. In practical terms, the result argues for controlled encoder adaptation rather than either fully fixed feature extraction or unconstrained source-task optimization.

The negative result for the MD-derived contact-persistence label is equally important. It shows that the failure mode is not a lack of simulation in general, but a mismatch between the source label and the target phenotype. Contact persistence in high-temperature trajectories may report aspects of structural durability, but it is a compressed trajectory statistic and may miss the mutation-specific energetic effects that determine how sequence changes alter the stability landscape. This distinction matters for future database design. A simulation campaign optimized for target transfer should not only maximize the number of labels; it should choose labels whose physical meaning is expected to transfer to the experimental decision being made.

The generated-variant result points toward a possible design workflow. Rosetta scores assigned to ESM2-generated variants improved T_m prediction relative to T_m-only learning, although generated variants were not conclusively better than random variants in the current comparison. This should be viewed as a bridge, not a completed design loop. The positive signal indicates that proposed sequences can be labeled by physics-based calculations and used as supervised source data for a T_m predictor. A natural next step is an iterative loop in which a generator proposes variants, physics-based calculations annotate them, the T_m predictor is updated, and candidates are selected for experimental testing. The present work supplies evidence for the supervised learning component of that loop, but it does not test

active learning, reinforcement learning, or experimental validation of newly proposed high-T_m nanobodies.

Limitations. Several limitations should guide interpretation. First, the experimental T_m training set is small, and the conclusions should be tested on additional nanobody collections and other protein families. Second, the source-label comparison is limited to the computational labels available here. Other free-energy protocols, stability predictors, trajectory summaries, or structure-based descriptors may transfer differently. Third, the FEP curve does not establish a clean power-law scaling relation; it supports a beneficial trend and a strong final improvement, but not a universal scaling exponent. Fourth, no new experimental measurements were performed to validate high-stability candidates. Finally, raw simulation trajectories may be too large to deposit fully, so reproducibility will depend on processed data, scripts, and clear reporting of the simulation protocol.

Outlook. The practical lesson is that simulation labels should be treated as design choices. For low-data experimental protein-property prediction, the relevant question is not whether simulations can produce many labels, but whether those labels encode information that improves the target task under a target-centered validation protocol. In this study, mutation free-energy labels provided that information for nanobody T_m prediction. This supports a broader strategy for biomolecular design: use experimental data to define the target phenotype, use simulation to annotate physically meaningful perturbations around that phenotype, and evaluate the combined model only on held-out experimental data. Such a strategy offers a disciplined path from simulation-assisted prediction to future closed-loop protein engineering.

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AUTHOR CONTRIBUTIONS

Taihei Murakami and Kentaro Sasaki curated data, performed computational experiments, analyzed results, and prepared figures. Yasuhiro Matsunaga conceived and supervised the study, acquired funding, and contributed to interpretation. All authors contributed to manuscript writing and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

Processed data required to reproduce the main and supplementary figures will be deposited in Zenodo at [ZENODO_DATA_DOI_PLACEHOLDER](#). Raw simulation trajectories are large and will be made avail-

able at [RAW_DATA_LOCATION_OR_REQUEST_POLICY_PLACEHOLDER](#).

Code availability

Analysis, training, and figure-generation code will be available on GitHub at [GITHUB_REPOSITORY_URL_PLACEHOLDER](#). An archival release will be deposited in Zenodo at [ZENODO_CODE_DOI_PLACEHOLDER](#).

AI-use disclosure

AI-assisted tools were used to support language editing, code development, and figure preparation. The authors reviewed and take responsibility for all manuscript content, analyses, and conclusions.

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Supplementary Note 1: Supplementary Information

Supplementary information will include extended source-label sweeps, additional encoder-size controls, hyperparameter search summaries, and tabulated paired error statistics supporting the main figures.

A. Supplementary Methods. The final supplementary methods will report the exact experimental splits, source-label preprocessing steps, training schedules, search grids, random seeds, bootstrap procedure, and software versions used to reproduce the reported comparisons.

B. Supplementary Figures and Tables. The final supplement will provide additional plots for development-set selection, source-label count controls, frozen-encoder controls, and model-size comparisons, together with machine-readable result tables.